# LA-WBC: Learning-driven Adaptation of Whole-Body Control for Safe Locomotion on Uncertain Terrain

Hanbin Jang and Jaemin Lee, *Member*, *IEEE*

***Abstract*—This paper proposes Learning-driven Adaptation of Whole-Body Control (LA-WBC) for quadrupedal locomotion on terrains with uncertain friction. LA-WBC infers terrain friction from proprioceptive feedback and WBC command–response features, and jointly adapts the friction coefficient used in the contact constraints and the commanded locomotion velocity. Unlike end-to-end reinforcement learning (RL), the proposed framework retains the nominal WBC structure and remains compatible with established WBC safety and stability analyses. Hardware-informed model matching and feature scaling improve sim-to-real friction inference and prediction consistency. Compared with the evaluated RL baseline, the proposed pipeline requires approximately 12% of the total data-generation and training time. Simulation and hardware experiments demonstrate improved velocity tracking, reduced foot slippage and base-motion variation, and higher locomotion success rates relative to non-adaptive WBC.**

***Index Terms*—Legged Robots, Whole-Body Motion Planning and Control, Motion Control**

## I. Introduction

RECENT advances in legged robotics have increased the demand for reliable locomotion over challenging and uncertain terrain, such as slippery surfaces. Model-based controllers perform well with accurate models and parameters but may degrade on such terrains without adaptation. End-to-end reinforcement learning (RL) offers adaptability but often requires extensive data and lacks explicit mechanisms for enforcing physical feasibility and safety. In contrast, whole-body control (WBC) provides a structured framework compatible with established safety and stability analyses, but typically relies on well-tuned parameters.

We propose Learning-driven Adaptation of WBC (**LA-WBC**), which adapts control parameters to terrain conditions without replacing the nominal WBC structure. Using the WBC for data generation and deployment preserves its model-based constraints and compatibility with existing safety and stability properties while introducing terrain adaptability. Based on friction inferred from proprioceptive feedback, LA-WBC jointly updates the WBC friction parameter and safely achievable locomotion velocity in real time.

Manuscript received: April 30, 2026; Revised: July 31, 2026; Accepted: September 27, 2026. This paper was recommended for publication by Editor Cosimo Della Santina upon evaluation of the Associate Editor and Reviewers' comments. The authors acknowledge support from the NVIDIA Academic Grant Program and Sequor Robotics for providing hardware used in part of the experiments.

Hanbin Jang and Jaemin Lee are with the Department of Mechanical and Aerospace Engineering, North Carolina State University, Raleigh, NC, USA (e-mail: hjang4@ncsu.edu; jaemin.lee@ncsu.edu).

Project website: https://2nitrogen.github.io/LA-WBC_project-page/

Digital Object Identifier (DOI): see top of this page.

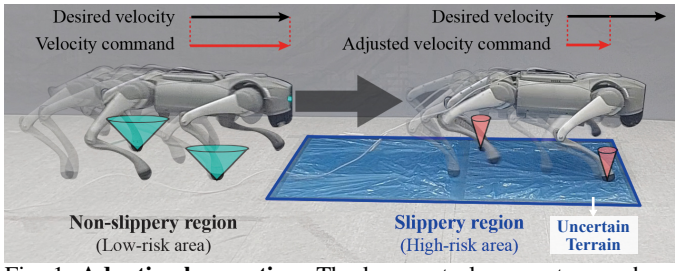



Fig. 1: **Adaptive locomotion:** The key control parameters such as contact wrench constraints and commanded velocities are adjusted based on the terrain-dependent admissibility.

### A. Related Works

Model-based motion planning and WBC methods are widely used for legged locomotion. WBC generally incorporates hierarchical task objectives and contact constraints through task-space optimization [1], together with gait-dependent trajectories and full-body dynamics for coordinated ground interaction [2], [3]. However, locomotion performance depends on accurate robot models and terrain properties. In particular, friction conditions, typically formulated as cone constraints parameterized by the Coulomb friction coefficient [4], directly affect the feasible locomotion velocity under full-body dynamic constraints [5].

Furthermore, model-based locomotion approaches can suffer from foot slippage or loss of balance due to model and parameter inaccuracies [6] and sim-to-real discrepancies [7]. Therefore, extensive trial and error has been required to tune parameters and adjust models. This has led to conservative strategies such as limiting locomotion speed, underestimating friction coefficients, or switching to more stable gait patterns [6], [8]. Consequently, these approaches have struggled to respond actively and effectively to the varying risks induced by environmental changes. As illustrated in Fig. 1, these strategies limit active responses to changing environmental risks, motivating autonomous control-parameter adaptation.

While incorporating accurate environmental properties thus significantly improves WBC-based locomotion, precisely estimating these properties remains a fundamental challenge. Inferring friction from measured contact forces requires foot-mounted force/torque (F/T) sensors, which produce noisy signals and are fragile under the repeated impacts of locomotion [9]. Moreover, many quadruped platforms do not include full 3-dimensional F/T sensors at the feet and instead rely primarily on pressure-based sensors, which provide only limited contact information and further restrict accurate force reconstruction. Exteroceptive vision provides predictive terrain information [10], [11], but can be unreliable on visually ambiguous terrain [12]. Visual friction estimation also depends

on surface and contact properties that may not be observable from appearance alone [13].

End-to-end reinforcement learning (RL) has demonstrated adaptive quadrupedal locomotion over diverse terrains [14], [15]. However, robust training under randomized friction coefficients and robot masses [16] requires substantial data and computation [17]. For example, [18] reports training times of 82 and 112 hours for gait-planning and control policies, respectively. Transferring these policies to new platforms or adapting them to changing physical conditions can also involve further fine-tuning or retraining because of limited generalization and safety concerns [19]–[21].

RL-based policies encode system dynamics, physical properties, and environmental parameters only implicitly, making it difficult to directly exploit inferred physical quantities during deployment. They may therefore require additional mechanisms to enforce feasibility, stability, and safety when integrated with model-based controllers such as WBC [22], [23]. To address these limitations, hybrid approaches have incorporated model-based optimal trajectories into RL training [24], [25], leveraged WBC to train policies for footstep planning, tracking, and recovery [26], and applied RL to online trajectory adaptation [27], footstep selection [28], and gait-mode selection [29]. However, these methods primarily adapt learned policies or motion decisions rather than directly updating physically interpretable WBC parameters.

Despite these advances, achieving safe and adaptive locomotion remains challenging, as illustrated by the sudden transition to high-risk terrain in Fig. 1. Rather than relying solely on a pretrained policy, we propose a learning-driven WBC adaptation framework that embeds adaptive intelligence within a model-based controller, bridging model-based reliability and learning-based adaptability. The framework retains the model-based structure of WBC and its compatibility with established safety and stability analyses while autonomously adapting the friction parameter and commanded velocity using proprioceptive feedback.

### B. Contributions

The primary contribution is a learning-driven WBC adaptation framework that uses WBC command–response features to infer terrain friction and jointly update the contact-friction parameter and commanded locomotion velocity while retaining the nominal controller. The detailed contributions of this paper are fourfold:

- LA-WBC infers terrain friction using onboard proprioceptive measurements and WBC command signals, without additional foot force/torque or vision sensors. It autonomously updates the WBC friction parameter and locomotion speed to reduce tracking error and maintain stable locomotion over slippery terrain.
- The proposed Recursive Graph Neural Network (RecGNN) is integrated with the existing WBC structure, while the complete data-generation and training pipeline requires only approximately $12\%$ of the total computational time of the evaluated end-to-end RL baseline.
- Four candidate NN architectures and an $\alpha$-scaled RecGNN variant are compared in terms of inference accuracy and computational cost, using both kinematic features and WBC command signals as inputs. The RecGNN achieves the best friction inference for slippery-terrain locomotion.
- A simulation-based calibration aligns the robot and contact models with the real system to reduce sim-to-real discrepancies, while the adaptability of the proposed framework is validated through simulation and hardware experiments.

## II. Preliminaries

This section presents a layered locomotion control framework based on WBC (Section II-A) and defines the specific problem addressed in this study (Section II-B).

### A. Layered Locomotion Control Architecture

Quadruped robots are commonly controlled through a model-based locomotion control framework that integrates footstep planning with WBC. Footstep planning, which is based on Raibert heuristics [30], determines subsequent footsteps from the desired body twist $\mathcal{V}^d = [\omega_z^d, v_x^d,\ v_y^d]^\top \in \mathbb{R}^3$. The corresponding joint position $\boldsymbol{q}^d \in \mathcal{Q} \subset \mathbb{R}^n$ and velocity $\dot{\boldsymbol{q}}^d \in T_q\mathcal{Q}$ references, computed by solving inverse kinematics problems, are tracked via WBC to realize the desired leg swing motions.

Given the terrain friction coefficient $\mu$, allowable joint acceleration $\mathcal{P} \subset \mathbb{R}^n$, and actuator torque limits $\mathcal{U} \subset \mathbb{R}^{n_u}$, where $n_u = n - 6$ denotes the actuation dimension, WBC formulates a constrained optimization problem, which is typically posed as a quadratic program (QP):

$$\begin{aligned} \min_{\boldsymbol{u},\, \Delta\ddot{\boldsymbol{q}},\, \mathbf{F}_c} \quad & \boldsymbol{u}^\top \mathbf{W}_u \boldsymbol{u} + \Delta\ddot{\boldsymbol{q}}^\top \mathbf{W}_\Delta \Delta\ddot{\boldsymbol{q}} + \mathbf{F}_c^\top \mathbf{W}_c \mathbf{F}_c, \\ \text{s.t.} \quad & \mathbf{D}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \mathbf{H}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \mathbf{B}\boldsymbol{u} + \mathbf{J}_c(\boldsymbol{q})^\top \mathbf{F}_c, \\ & \Delta\ddot{\boldsymbol{q}} = \ddot{\boldsymbol{q}} - \ddot{\boldsymbol{q}}^d, \quad \ddot{\boldsymbol{q}} \in \mathcal{P} \\ & \mathbf{J}_s(\boldsymbol{q})\Delta\ddot{\boldsymbol{q}} = -\dot{\mathbf{J}}_s(\boldsymbol{q}, \dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} \;-\; \mathbf{J}_s(\boldsymbol{q})\,\ddot{\boldsymbol{q}}^d, \\ & \mathbf{F}_c \in \mathcal{CWC}(\mu, \boldsymbol{q}), \quad \boldsymbol{u} \in \mathcal{U} \end{aligned} \tag{1}$$

where $\mathbf{D}(\boldsymbol{q}) \in \mathbb{S}^n_{>0}$, $\mathbf{H}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^n$, and $\mathbf{B} \in \mathbb{R}^{n \times n_u}$ denote the mass/inertia matrix, the sum of Coriolis, centrifugal, and gravitational forces, and the selection matrix for the actuation, respectively. $\mathbf{J}_c(\boldsymbol{q}) \in \mathbb{R}^{n_c \times n}$ and $\mathbf{J}_s(\boldsymbol{q}) \in \mathbb{R}^{n_s \times n}$ are the stacked contact Jacobian and the Jacobian associated with stance contacts where $n_c$ and $n_s$ are the dimensions of the full stacked contact force and stance-foot contact force. $\mathbf{W}_u \in \mathbb{R}^{n_u \times n_u}$, $\mathbf{W}_\Delta \in \mathbb{R}^{n \times n}$, and $\mathbf{W}_c \in \mathbb{R}^{n_c \times n_c}$ are diagonal weighting matrices corresponding to the quadratic cost terms with control torque $\boldsymbol{u}$, joint acceleration deviation $\Delta\ddot{\boldsymbol{q}}$, and contact force $\mathbf{F}_c$. $\mathcal{CWC}(\mu, \boldsymbol{q})$ denotes the contact wrench cone parameterized by $\mu$ and $\boldsymbol{q}$ [4].

Within the WBC, a safety filter modifies the desired base acceleration to enforce prescribed body-velocity limits:

$$\ddot{\boldsymbol{q}}^d_{\text{base}} = \arg\min_{\ddot{\boldsymbol{q}}_{\text{base}}} \|\ddot{\boldsymbol{q}}_{\text{base}} - \ddot{\boldsymbol{q}}^d_{\text{base}}\|^2, \quad \ddot{\boldsymbol{q}}_{\text{base}} \in \mathcal{P}_{\text{base}} \tag{2}$$

$$\text{s.t.} -\ddot{\boldsymbol{q}}_{\text{base}} + \xi(\mathcal{V}_{\max} - \mathcal{V}) \succeq \mathbf{0},\ \ddot{\boldsymbol{q}}_{\text{base}} + \xi(\mathcal{V} - \mathcal{V}_{\min}) \succeq \mathbf{0}$$

where $\ddot{\boldsymbol{q}}_{\text{base}} \in \mathbb{R}^3$ contains the $x$, $y$, and yaw accelerations, and $\ddot{\boldsymbol{q}}^d_{\text{base}}$ is generated from the command $\mathcal{V}^d$ through feedback PD control. The measured body twist is denoted by $\mathcal{V} = [\omega_z, v_x,\ v_y]^\top \in \mathbb{R}^3$. The bounds $\mathcal{V}_{\max}$ and $\mathcal{V}_{\min}$ define the admissible body velocities, with $\xi(h) = c_\xi h$ and $c_\xi > 0$. As a supervisory fail-safe, the controller switches to a protective

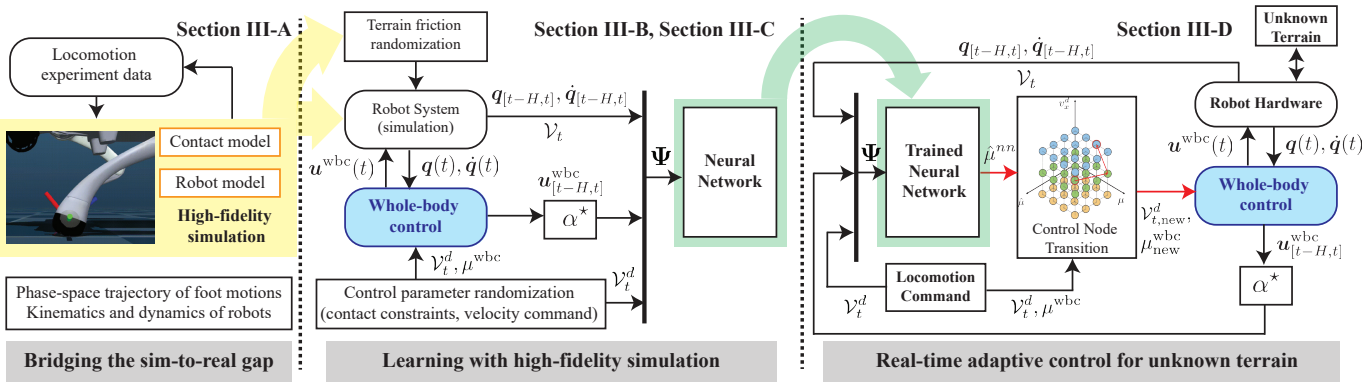



Fig. 2: **Overall architecture:** Using simulation, we train an NN that infers the terrain friction coefficient through supervised learning. Based on the inferred friction coefficient, the robot adapts its locomotion behavior in response to terrain properties.

lie-down mode when the acceleration correction exceeds a predefined threshold and the foot contact is unstable.

After obtaining the optimal decision variables $\boldsymbol{u}^\star$, $\Delta\ddot{\boldsymbol{q}}^\star$, $\mathbf{F}_c^\star$, the desired joint positions and velocities are updated via Euler integration using the joint accelerations $\ddot{\boldsymbol{q}} = \ddot{\boldsymbol{q}}^d + \Delta\ddot{\boldsymbol{q}}^\star$. The resulting feedforward torque command is then applied through the joint impedance controller:

$$\boldsymbol{u}^{\text{wbc}} = \mathbf{K}_p(\boldsymbol{q}^d - \boldsymbol{q}) + \mathbf{K}_d(\dot{\boldsymbol{q}}^d - \dot{\boldsymbol{q}}) + \boldsymbol{u}^\star \tag{3}$$

where $\mathbf{K}_p$ and $\mathbf{K}_d$ represent the proportional and derivative gain matrices. Under fixed contact, exact feedback linearization, and feasible reference tracking, these gains render the nominal tracking-error dynamics exponentially stable [31], while the joint-space impedance controller provides dissipative, passive-like behavior [32], [33]. The WBC operates at an update rate on the order of 1 kHz, comparable to prior implementations running at 1.5 kHz [3], enabling reliable real-time locomotion control.

### B. Problem Statement

**Assumption**: A quadruped robot is equipped only with pressure sensors and therefore cannot reliably measure three-dimensional contact forces. A nominal WBC parameter set has been identified that enables reliable locomotion when the friction coefficient is accurately specified.

**Problem**: Safe and balanced locomotion requires the ground reaction forces to remain within the contact wrench cone, $\mathbf{F}_c \in \mathcal{CWC}(\mu, \boldsymbol{q})$. However, WBC computes contact forces subject to constraints parameterized by a modeled friction coefficient $\mu^{\text{wbc}}$, which may differ from the actual terrain coefficient $\mu$. This mismatch, $\mu^{\text{wbc}} \neq \mu$, can produce contact forces that violate the actual contact wrench cone, thereby degrading locomotion performance and potentially causing loss of balance. Moreover, terrain friction constrains the range of safely achievable walking velocities; therefore, adapting $\mu^{\text{wbc}}$ alone is insufficient.

We therefore define the terrain-dependent admissible set $\mathcal{C}_{\text{adm}}(\mu) = \{(\mu^{\text{wbc}}, \mathcal{V}^d) : \mathcal{V}^d \in \mathcal{S}(\mu^{\text{wbc}}, \mu)\}$ where $\mathcal{S}(\mu^{\text{wbc}}, \mu)$ denotes the set of velocity commands safely achievable by WBC under the controller parameter $\mu^{\text{wbc}}$ when executing on real-world terrain with friction coefficient $\mu$. We define the problem as designing a real-time adaptation mechanism that estimates the terrain-dependent admissible set and jointly adjusts $(\mu^{\text{wbc}}, \mathcal{V}^d)$ under varying terrain conditions to maintain safe and feasible locomotion.

## III. The Proposed Approach

The proposed framework consists of four stages as illustrated in Fig. 2. First, the robot and contact models are calibrated using hardware trajectories (Section III-A). Second, WBC rollouts are collected while varying terrain friction, the WBC friction parameter, and the velocity command (Section III-B). Third, the trained NN maps histories of proprioceptive and WBC command signals to a continuous friction inference $\hat{\mu}^{nn}$ (Section III-C). Finally, a Hidden Markov Model (HMM) converts this inference into a discrete terrain state, based on which the friction parameter and commanded velocity are adapted (Section III-D).

### A. Sim-to-Real Gap Mitigation

As illustrated in Fig. 2, two sources of discrepancy are considered: the robot model and the contact model. To address the kinematic mismatch in the robot model, we adjust the foot frames by comparing hardware measurements with simulation data. Adjusting the contact model is more challenging, as the high-fidelity simulator defines five key parameters that govern contact behavior [34]. To reduce the sim-to-real gap in the contact model, we conduct in-place stepping experiments on a high-friction terrain. Following [35], the friction coefficient was measured using a dummy mass, the robot foot, and a push-pull gauge. Under the Coulomb friction assumption on flat surfaces, the dummy mass was set to approximately one-quarter of the robot's total weight. The five key contact parameters were then identified by minimizing the root-mean-square error (RMSE) between measured trajectories and those generated by the constraint force model in [34]. We refer to this sim-to-real gap reduction procedure as hardware-informed model matching.

Hardware-informed model matching improves the physical fidelity and dynamic consistency of the collected datasets. As shown in Fig. 3, substantial discrepancies remain between hardware (red) and simulation (blue) trajectories even with the same friction coefficient, which can significantly degrade NN prediction accuracy during real-world deployment. The proposed method reduces this sim-to-real gap, yielding more consistent phase-space trajectories (green).

### B. Dataset Construction

We collect proprioceptive signals from a high-fidelity simulation in which locomotion is controlled by the model-based WBC. Each feature vector combines joint positions

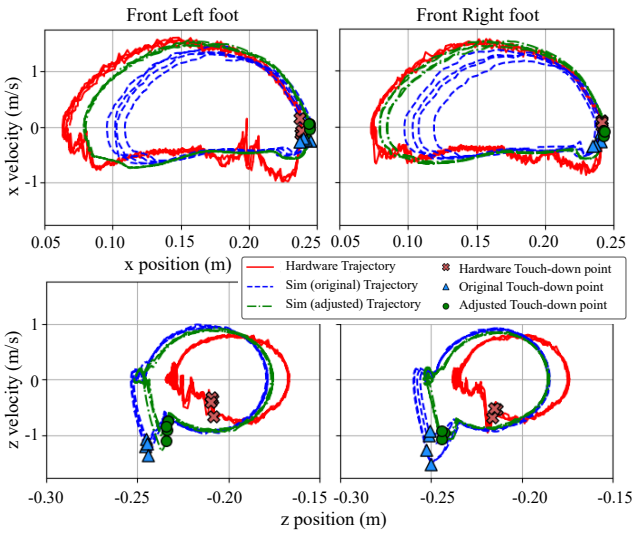



Fig. 3: **Sim-to-real gap**: Phase-space trajectories of the front feet.

and velocities, WBC control inputs, and desired and measured base twists. The desired and measured twists with the WBC commands capture tracking deviations caused by slippage. The stacked feature vector at time $t$ is defined as follows:

$$\boldsymbol{\Psi}_t = [\underbrace{\boldsymbol{q}_t^\top,\ \dot{\boldsymbol{q}}_t^\top,\ (\mathcal{V}_t^d)^\top,\ \mathcal{V}_t^\top}_{\text{kinematic}},\ \underbrace{(\boldsymbol{u}_t^{\text{wbc}})^\top}_{\text{dynamic}}]^\top \in \mathbb{R}^{6+3n_u}. \quad (4)$$

The NN input, $\boldsymbol{\Psi}_{[t-H:t]}$, is constructed from the history of the finite feature vector $\boldsymbol{\Psi}$ over a time horizon $[t-H:t]$.

The dataset is generated by varying environment and controller parameters across episodes. We define the corresponding parameter tuples as $\boldsymbol{\Omega}^e := (\mu^s)$ and $\boldsymbol{\Omega}^c := (\mu^{\text{wbc}}, \mathcal{V}^d)$ where $\mu^s$ is the friction coefficient in the simulation environments. This formulation focuses on learning their relationship but can be extended to additional variables, such as terrain slope and gait pattern. During data collection, we sweep the combined parameter tuple $\boldsymbol{\Omega}_i$ for all $i \in \{1, \dots, N_\Omega\}$ where the total number of combinations is $N_\Omega = N_{\mu^s} N_{\mu^{\text{wbc}}} N_{\mathcal{V}^d}$. Each term denotes the number of sampled values or vectors for the corresponding parameter. Because the dataset size grows multiplicatively with sampling resolution, the sampling ranges and densities are selected to balance coverage of physically meaningful operating conditions and data-collection cost.

### C. Training Neural Networks

In this section, we describe the training procedure for NNs used to infer terrain friction coefficients within the WBC framework, as shown in the second step of Fig. 2. We then present the additional scaling applied after z-standardization to improve feature distinguishability, followed by the proposed network architectures and their training procedures.

**Data Preprocessing**: The NN input $\boldsymbol{\Psi}_{[t-H:t]}$ consists of heterogeneous physical information with distinct levels and sensitivities, which can lead to over or under-representation of certain components during training. To mitigate this bias, we introduce a feature-wise scaling strategy that better separates the characteristics of each input modality. Specifically, we re-index the input as $\boldsymbol{\Psi} = [\psi_1^\top, \psi_2^\top]^\top$ with kinematic, $\psi_1 \in \mathbb{R}^{n_{\psi_1}}$, and dynamic, $\psi_2 \in \mathbb{R}^{n_{\psi_2}}$, components. Based on this decomposition, we compute the covariance: $\boldsymbol{\Sigma}_\Psi = \mathbb{E}[(\boldsymbol{\Psi} - \overline{\boldsymbol{\Psi}})(\boldsymbol{\Psi} - \overline{\boldsymbol{\Psi}})^\top]$ where $\overline{\boldsymbol{\Psi}} = \mathbb{E}[\boldsymbol{\Psi}]$. Our approach introduces a weighting factor $\alpha$ for the dynamic components to compensate for their relative

**Algorithm 1** RecGNN for Terrain Parameter Inference

**Given:** Base/joint-state histories $\mathbf{x}_t^b \in \mathbb{R}^6$, $\mathbf{x}_{t,\ell}^j \in \mathbb{R}^9$ ($t = 1, \dots, H$, $\ell = 1, \dots, 4$); message-passing steps $K$

**for** $t = 1$ to $H$ **do**
  $\mathbf{n}_{t,0}^{(0)} \leftarrow \zeta_{\text{base}}(\mathbf{x}_t^b)$, $\mathbf{n}_{t,\ell}^{(0)} \leftarrow \zeta_{\text{leg}}(\mathbf{x}_{t,\ell}^j)$
  **for** $k = 1$ to $K$ **do**
    $\mathbf{m}_{t,i\to j}^{(k)} \leftarrow \zeta_{\text{edge}}(\mathbf{n}_{t,i}^{(k-1)}, \mathbf{n}_{t,j}^{(k-1)})$, $(i,j) \in \mathcal{E}$
    $\mathbf{n}_{t,i}^{(k)} \leftarrow \zeta_{\text{node}}(\mathbf{n}_{t,i}^{(k-1)}, \sum_{j\in\mathcal{N}(i)} \mathbf{m}_{t,j\to i}^{(k)})$
  **end for**
  $\mathbf{s}_t \leftarrow \text{Concat}[\mathbf{n}_{t,0}^{(K)}, \dots, \mathbf{n}_{t,4}^{(K)}]$
**end for**
$\mathbf{h}_H \leftarrow \text{GRU}(\{\mathbf{s}_t\}_{t=1}^H)$
$\hat{\mu}^{nn} \leftarrow \zeta_{\text{head}}(\mathbf{h}_H)$

sensitivity. Accordingly, the covariance matrix is modified as follows:

$$\boldsymbol{\Sigma}_\Psi^\alpha = \begin{bmatrix} \mathbf{1}_{n_{\psi_1}\times n_{\psi_1}} & \alpha \mathbf{1}_{n_{\psi_1}\times n_{\psi_2}} \\ \alpha \mathbf{1}_{n_{\psi_2}\times n_{\psi_1}} & \alpha^2 \mathbf{1}_{n_{\psi_2}\times n_{\psi_2}} \end{bmatrix} \odot \boldsymbol{\Sigma}_\Psi \quad (5)$$

where $\odot$ denotes the Hadamard product operator and $\mathbf{1}_{n_a\times n_b}$ is an all-ones matrix of size $n_a \times n_b$.

The weighting factor $\alpha$ is selected to improve the numerical conditioning of the input covariance. Specifically, we evaluate $\kappa(\boldsymbol{\Sigma}) = \frac{\lambda_{\max}(\boldsymbol{\Sigma})}{\lambda_{\min}(\boldsymbol{\Sigma})}$ where $\lambda(\boldsymbol{\Sigma})$ denotes an eigenvalue of the covariance matrix $\boldsymbol{\Sigma}$. A smaller condition number indicates a better-balanced feature representation and reduces disparities in the learning dynamics across input dimensions. To obtain a more appropriate value of $\alpha$, we establish a lower bound on the condition number as follows:

$$\begin{aligned} \|\boldsymbol{\Sigma}\|_{F/2}^2 &:= \frac{\|\boldsymbol{\Sigma}\|_F^2}{\|\boldsymbol{\Sigma}\|_2^2} = \frac{\sum_{i=1}^{n_\sigma} \lambda_i^2(\boldsymbol{\Sigma})}{\lambda_{\max}^2(\boldsymbol{\Sigma})} \geq 1 + \frac{n_\sigma - 1}{\kappa(\boldsymbol{\Sigma})^2}, \\ \kappa(\boldsymbol{\Sigma}) &\geq \sqrt{(n_\sigma - 1)(\|\boldsymbol{\Sigma}\|_{F/2}^2 - 1)^{-1}}. \end{aligned} \quad (6)$$

where $n_\sigma$ is the dimension of $\boldsymbol{\Sigma}$. In addition, $\|\cdot\|_2$ and $\|\cdot\|_F$ denote the matrix 2-norm and Frobenius-norm, respectively.

**Remark 1.** To identify candidate scaling factors that may improve the conditioning of the weighted covariance matrix, we retain values of $\alpha$ satisfying $\|\boldsymbol{\Sigma}_\Psi^\alpha\|_{F/2}^2 \geq \|\boldsymbol{\Sigma}_\Psi\|_{F/2}^2$. This condition increases the stable-rank-based measure of the weighted covariance matrix and thereby reduces the corresponding lower-bound surrogate for its condition number.

After collecting all $\alpha$ values that satisfy the condition in Remark 1 into a set $\mathcal{A}$, we solve the following optimization problem to determine the optimal scaling parameter that maximizes feature distinguishability:

$$\begin{aligned} \alpha^\star = \arg\max_{\alpha\in\mathcal{A}} \quad & \|\kappa(\boldsymbol{\Sigma}_\Psi) - \kappa(\boldsymbol{\Sigma}_\Psi^\alpha)\|_2^2 \\ \text{s.t.} \quad & \kappa(\boldsymbol{\Sigma}_\Psi) - \kappa(\boldsymbol{\Sigma}_\Psi^\alpha) \geq 0, \quad \alpha > 0. \end{aligned} \quad (7)$$

In this paper, because $\mathcal{A}$ is constructed as a finite candidate set, $\alpha^\star$ is obtained through a sampling-based search.

**Neural Network Training**: We train the NNs in a supervised manner using the simulated terrain friction coefficient $\mu^s$ of each episode as the ground-truth label. Because network architectures exhibit different inductive biases, we compare four models: 1) a fully connected neural network (FCNN), 2) a gated recurrent unit (GRU), 3) a transformer, and 4) the proposed recursive graph neural network (RecGNN).

**RecGNN**: The proposed RecGNN is a spatiotemporal GNN [36] inspired by quadruped morphology. Its graph layers perform message passing over the robot's physical topology to capture instantaneous interactions between the base and legs, while a GRU recursively updates a hidden state over the input history to capture their evolution over time for terrain-friction inference. As detailed in Algorithm 1, the graph follows a star topology in which the base node, $i = 0$, is connected bidirectionally to the four leg nodes, $i = 1, \ldots, 4$. Accordingly, the edge set is $\mathcal{E} = \{(0, \ell), (\ell, 0) \mid \ell = 1, \ldots, 4\}$ with neighborhoods $\mathcal{N}(0) = \{1, 2, 3, 4\}$ and $\mathcal{N}(\ell) = \{0\}$, and $\zeta_{(\cdot)}$ denotes a task-specific multilayer perceptron.

### *D. Online Adaptive Strategy*

This section presents an online scheme that jointly updates the WBC friction parameter and achievable locomotion velocity based on $\hat{\mu}^{nn}$.

**HMM**: The HMM is employed to estimate the most probable discrete terrain state $\hat{\mu}_{\pi}^{nn}$ from the continuous prediction $\hat{\mu}^{nn}$. We represent $\mathbf{\Omega}_i^e$ as the $i$-th discretized terrain state. We define the transition function $\boldsymbol{T}(\mathbf{\Omega}_i^e, \mathbf{\Omega}_j^e) = \mathbb{P}(\mathbf{\Omega}_t^e = \mathbf{\Omega}_i^e \mid \mathbf{\Omega}_{t-1}^e = \mathbf{\Omega}_j^e)$ and model it under the Poisson assumption:

$$\mathbf{T}(\mathbf{\Omega}_i^e, \mathbf{\Omega}_j^e) = \begin{cases} \gamma & \text{if } i = j \\ \frac{1-\gamma}{N_{\Omega^e}-1} & \text{else.} \end{cases} \tag{8}$$

where $\gamma = \exp(-\rho)$ is the probability of remaining in the same state. The terrain-changing intensity is $\rho \lesssim v_{x,\max}^d \, dt/L$, where $v_{x,\max}^d$ denotes the maximum commanded forward velocity in the predefined domain $\mathbf{\Omega}^c$, and $L$ is the characteristic support length along the dominant direction of motion. This approximation is appropriate for the planar, friction-varying terrains and moderate forward velocities considered in this study.

The observation likelihood is formulated as follows: $\ell_t(i) = \mathcal{N}\left(\hat{\mathbf{\Omega}}_t^e \,\middle|\, \mathbf{\Omega}_i^e, \mathbf{\Sigma}_i\right)$ for all $i \in [1, N_{\Omega^e}]$ where $\hat{\mathbf{\Omega}}_t^e$ denotes the prediction of $\mathbf{\Omega}_t^e$. In addition, $\mathbf{\Sigma}_i$ is estimated from state-wise prediction residuals on a held-out dataset. Given the observation likelihood, the terrain-state belief is updated as follows:

$$\boldsymbol{\pi}_t(i) = \frac{\ell_t(i) \sum_j \mathbf{T}(\mathbf{\Omega}_i^e, \mathbf{\Omega}_j^e) \boldsymbol{\pi}_{t-1}(j)}{\sum_k \ell_t(k) \sum_j \mathbf{T}(\mathbf{\Omega}_k^e, \mathbf{\Omega}_j^e) \boldsymbol{\pi}_{t-1}(j)}. \tag{9}$$

Thus, the most probable terrain state is estimated as $\mathbf{\Omega}_{\hat{i}_t}^e$ by finding $\hat{i}_t = \arg\max_i \boldsymbol{\pi}_t(i)$.

**Offline Characterization of Locomotion Costs**: To characterize the velocity-tracking performance, energy efficiency, and stability-related base motion of each locomotion episode, we define the composite cost $\mathcal{L} = c_1\mathcal{L}_{\text{vel}} + c_2\mathcal{L}_{\text{cot}} + c_3\mathcal{L}_{\text{stab}}$ inspired by a prior quadrupedal locomotion framework [37], where $c_k$ denotes the $k$-th weighting factor. The three terms quantify velocity-tracking error, cost of transport, and periodicity-aware base-motion deviation, respectively:

$$\mathcal{L}_{\text{vel}} = \frac{\|\mathcal{V}^d - \overline{\mathcal{V}}_{[t_1,t_N]}\|_2}{\|\mathcal{V}^d\|_2}, \quad \mathcal{L}_{\text{cot}} = \frac{\sum_{i=1}^N \max(\boldsymbol{u}_{t_i}^\top \dot{\boldsymbol{q}}_{t_i}, 0)}{mg \sum_{i=1}^N \|\boldsymbol{v}_{t_i}^{xy}\|_2}$$

$$\mathcal{L}_{\text{stab}} = \frac{1}{N - 2H + 1} \sum_{i=2H}^{N} \exp\left[\Delta\theta_{t_i} + \beta\big(1 - r_{t_i}\big)^2\right] \tag{10}$$

Here, $\overline{\mathcal{V}}_{[t_i,t_f]}$ denotes the mean of base twist $\mathcal{V}$ over the interval $[t_i, t_f]$ and $\boldsymbol{v}_t^{xy}$ represents the linear velocity in the x–y plane at time $t$. In addition, $m$ and $g$ represent the total mass and gravitational acceleration, respectively. The quantity $\Delta\theta_t$ is the geodesic distance between consecutive base orientations. The coefficient $r_t$ is the Pearson correlation between the base orientation deviation over the adjacent windows $[t-H, t]$ and $[t-2H, t-H]$. Consequently, repeatable gait-induced oscillations receive a smaller penalty than abrupt or aperiodic base motions. The parameter $\beta > 0$ balances the angular-deviation and nonperiodicity terms. In this paper, a sensitivity analysis with respect to $\beta$ showed that the cost-map ranking remained largely unchanged over $\beta \in [0.25, 2.0]$, with Spearman rank correlations of at least $0.954$ relative to the nominal value $\beta = 0.5$.

**Gradient-based Update Method**: Based on the precomputed offline cost $\mathcal{L}(\mathbf{\Omega}_i^e, \mathbf{\Omega}_j^c)$, at time $t$, the estimate $\mathbf{\Omega}_{\hat{i}_t}^e$ selects the corresponding cost-map slice: $\mathcal{L}_t(\mathbf{\Omega}^c) = \mathcal{L}(\mathbf{\Omega}_{\hat{i}_t}^e, \mathbf{\Omega}^c)$. Thus, the environmental state is fixed through online lookup, while the WBC friction parameter and commanded velocity remain the independent variables. Over all possible values of $\mathbf{\Omega}^c$, we minimize the cost function $\mathcal{L}_t$ by updating the control parameters from the previous $\mathbf{\Omega}_{t-1}^c$ using a gradient descent method. To enable smooth gradient-based updates, we construct a locally smoothed approximation of the cost landscape $\mathcal{L}_t$ via linear interpolation and Gaussian filtering with a standard deviation $\sigma$. The detailed update rule is described as follows:

$$\widetilde{\mathbf{\Omega}}_t^c = \widetilde{\mathbf{\Omega}}_{t-1}^c - \eta \nabla_{\widetilde{\mathbf{\Omega}}^c} \mathcal{L}_t + \eta_g \left(\mathbf{\Omega}_t^{c\star} - \widetilde{\mathbf{\Omega}}_{t-1}^c\right) \tag{11}$$

where $\eta > 0$ is the step size of the gradient-descent term and $\eta_g > 0$ is the weight of the attraction toward the global minimizer $\Omega_t^{c\star}$ of $\mathcal{L}_t$ given the estimated environment tuple. The continuous state $\widetilde{\mathbf{\Omega}}_t^c$ is then projected onto the nearest discrete control parameter node for execution. The attraction term biases the update toward the globally lowest-cost node while the gradient term provides gradual local adaptation. The objective of this update is to keep the control parameters within the admissible set. The precomputed cost landscape serves as an empirical surrogate for locomotion performance under each terrain–controller combination. Accordingly, minimizing $\mathcal{L}_t$ steers the control parameters toward low-cost regions associated with favorable locomotion.

## IV. DEMONSTRATIONS

This section evaluates the proposed framework through complementary simulation and hardware experiments. Section IV-A presents the simulation and experimental setup, and Section IV-B describes parameter selection, training configuration, and computational cost. Sections IV-C and IV-D analyze sim-to-real friction inference and real-time locomotion adaptation, respectively.

### *A. Simulation and Experimental Setup*

We evaluate the framework on high- and low-friction planar terrains with friction coefficients of $0.76$ and $0.21$, respectively, measured as in Section III-A. These values are used only as evaluation ground truth and are not provided to either the friction estimator or the controller.

Hardware experiments were conducted at multiple commanded velocities, with representative results reported at $0.5$ and $0.7$ m/s for clear comparison. The WBC runs at

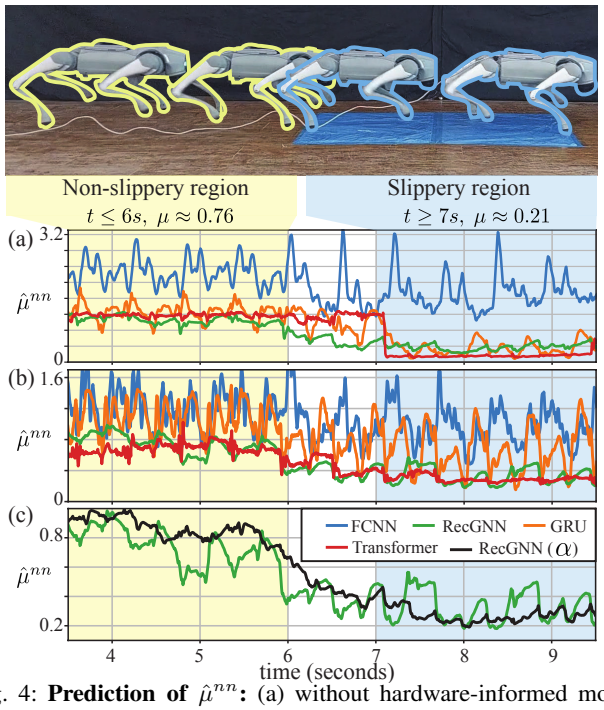



Fig. 4: **Prediction of $\hat{\mu}^{nn}$:** (a) without hardware-informed model matching, (b) with hardware-informed model matching, and (c) with $\alpha$-scaling.

1 kHz, while the complete adaptation pipeline, including friction inference, HMM-based terrain-state estimation, and control-parameter adaptation, operates at 200 Hz. All training, simulation, and hardware experiments were performed on a desktop workstation equipped with a 32-core x86_64 CPU and an NVIDIA RTX 4090 GPU.

### *B. Parameters, Training Setup, and Computation Time*

Table I summarizes the parameter settings and computational cost of the proposed method. In particular, $c_3$ was selected to penalize abrupt and aperiodic body motions without inducing unnecessarily conservative adaptation. The step sizes $\eta$ and $\eta_g$ were selected empirically to balance adaptation responsiveness and smooth parameter evolution. The history length $H$ was set to cover one complete 0.6 s trot cycle generated by the gait planner.

The training dataset was generated in MuJoCo over friction coefficients of 0.1–1.3 and commanded velocities of 0.1–0.7 m/s, both at a resolution of 0.1, yielding approximately 120 million time-step samples. The parameter $\alpha$ was selected using a randomly sampled 1% subset of the training data. The RecGNN uses 48-dimensional node embeddings and messages, an 80-dimensional MLP hidden width, a single-layer GRU with 48 hidden units, and one message-passing step per time step ($K = 1$). All candidate models were matched to comparable parameter counts and trained for 30 epochs using Adam with a learning rate and weight decay of $10^{-4}$, cosine annealing, and a batch size of 256.

For comparison, an end-to-end RL locomotion policy was trained using the public implementation in [38], with its velocity-command and friction-randomization ranges matched to those of our framework. The RL baseline required approximately 4.915 billion time-step samples and 29 h of total training time, whereas our complete data-generation, preprocessing, and training pipeline required 120 million samples and 3.53 h.

TABLE I. Parameter Settings and Computational Cost

| Params | $\gamma$ | $\beta$ | $c_1$ | $c_2$ | $c_3$ | $\sigma$ | $\eta$ | $\eta_g$ | $H$ | $\alpha^\star$ |
|---|---|---|---|---|---|---|---|---|---|---|
| (ours) | 0.995 | 0.5 | 1 | 1 | 0.2 | 1 | 0.02 | 0.05 | 120 | 0.37 |

| | Time Steps | Data Collection | Preprocessing | Training | Total Time |
|---|---|---|---|---|---|
| Vanilla RL | 4,915 M | 29 hrs (combined, 50k iterations) | | | 29 hrs |
| Ours | 120 M | 2.63 hrs | 0.5 hrs | 0.4 hrs | **3.53** hrs |

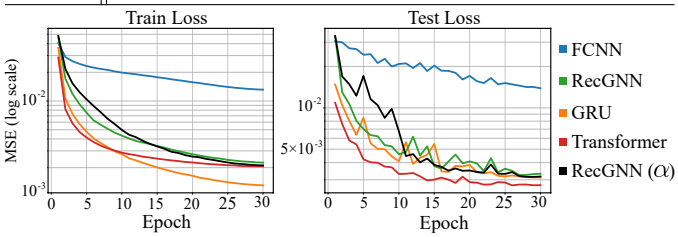



Fig. 5: **Loss trajectories.** (Left) Training loss for different NN architectures. (Right) Test loss evaluated during training.

Thus, the proposed method used only 2.4% of the samples and approximately 12% of the total computational time required by the RL baseline.

### *C. Sim-to-Real Inference Analysis*

We evaluate the sim-to-real friction-inference performance by comparing prediction MAE, standard deviation, and training and test losses. Fig. 4 compares four candidate architectures and the $\alpha$-scaled RecGNN variant during hardware locomotion. The comparison between Fig. 4(a) and Fig. 4(b) shows that hardware-informed model matching improves the prediction accuracy of all architectures, with RecGNN and Transformer outperforming FCNN and GRU. Fig. 4 further evaluates the proposed $\alpha$-scaling for RecGNN, which reduces oscillations while maintaining a low mean error on slippery terrain. This improvement is further confirmed by the prediction MAE and standard deviation presented in Table II. Fig. 5 also shows that FCNN converges more slowly and exhibits higher training and test losses. Overall, Transformer and RecGNN achieve the strongest inference performance, while the scaled RecGNN provides the most consistent predictions on slippery terrain and is therefore adopted in the complete framework.

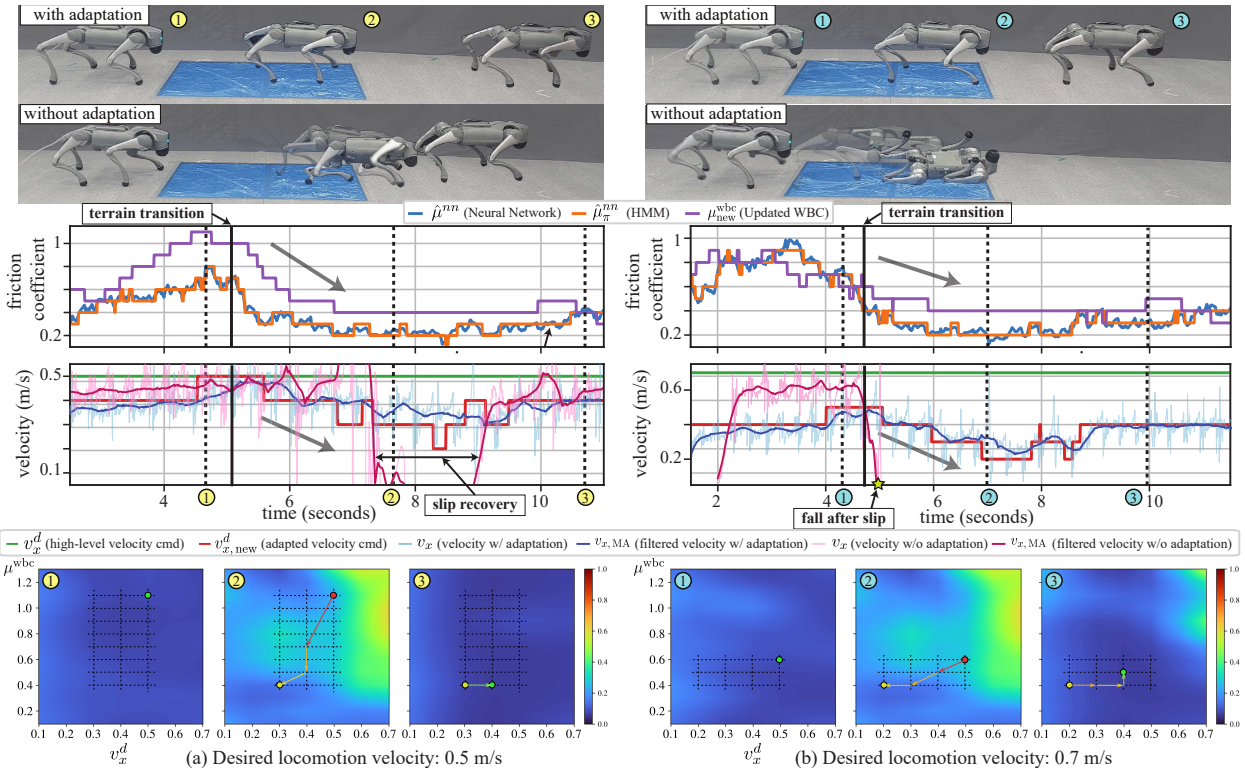
### *D. Online Locomotion Adaptation Analysis*

We evaluate the proposed framework through simulation and hardware experiments. In simulation, we compare the proposed adaptation method with non-adaptive WBC and the vanilla RL baseline, whereas the hardware experiments compare locomotion with and without adaptation. Table II includes the cost of transport (CoT), velocity tracking error (RMSE relative to each method's applied command), mean velocity (until failure or completion), stance-foot velocity, base-height deviation, and success rate (completion of a 10 m course containing at least 5 m of low-friction terrain). The velocity tracking error is computed with respect to the final velocity command, whereas the mean velocity represents the overall locomotion speed. The metrics are averaged over episodes of 9.3 s and 10.4 s for the 0.5 m/s and 0.7 m/s cases, respectively. For each method and velocity condition, 100 simulation trials were conducted with randomized initial robot positions to vary the gait phase and limb-contact state at which the robot entered the slippery terrain.

In the repeated simulation trials, the proposed method improves most performance metrics relative to non-adaptive WBC, with the exception of CoT. The representative hardware

TABLE II. Inference/prediction performance across different neural network models and locomotion performance.

| **Inference and Prediction Performance** | | | | | | **Locomotion Performance: Simulation (Hardware)** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Latency ($\mu$s) | | Prediction MAE | | | $v_x^d = 0.5$ m/s | | | $v_x^d = 0.7$ m/s | |
| Model | Params (M) | GPU | CPU | Non-slip ($\mu \approx 0.76$) | Slip ($\mu \approx 0.21$) | Metric | No-adapt | Vanilla RL | Ours | No-adapt | Vanilla RL | Ours |
| FCNN | 0.076 | $65 \pm 3$ | $22 \pm 6$ | $0.427 \pm 0.260$ | $0.768 \pm 0.244$ | Cost of Transport | 0.203 (0.326) | **0.153** | 0.299 (0.221) | 0.406 (0.225) | **0.136** | 0.297 (0.224) |
| GRU | 0.065 | $127 \pm 4$ | $890 \pm 13$ | $0.337 \pm 0.220$ | $0.466 \pm 0.281$ | Velocity tracking error (m/s) | 0.105 (0.306) | 0.180 | **0.070** (0.112) | 0.257 (0.591) | 0.212 | **0.070** (0.111) |
| Transformer | 0.075 | $305 \pm 10$ | $439 \pm 8$ | $\mathbf{0.093 \pm 0.079}$ | $0.088 \pm 0.045$ | Mean velocity (m/s) | **0.552** (0.444) | 0.332 | 0.319 (0.363) | **0.614** (0.571) | 0.504 | 0.318 (0.354) |
| RecGNN | 0.078 | $297 \pm 7$ | $870 \pm 13$ | $0.104 \pm 0.133$ | $0.125 \pm 0.107$ | Stance-foot velocity (m/s) | 0.297 (0.109) | 0.219 | **0.165** (0.087) | 0.354 (0.162) | 0.314 | **0.164** (0.089) |
| RecGNN ($\alpha$) | same as RecGNN | | | $0.107 \pm 0.077$ | $\mathbf{0.065 \pm 0.053}$ | Base-height std. (mm) | 2.108 (47) | 2.308 | **1.531** (3) | 38.419 (39) | 2.325 | **1.532** (3) |
| | | | | | | Success rate (%), 100 trials | 98 | **100** | **100** | 1 | **100** | **100** |




(a) Desired locomotion velocity: 0.5 m/s (b) Desired locomotion velocity: 0.7 m/s

Fig. 6: **Hardware experiments for adaptive locomotion**: The first, second, and third rows illustrate the experiment snapshots, the trajectories of control parameters, and the control parameter adjustments induced by the cost updates, with the adjusted values denoted by the subscript $(.)_{\text{new}}$. Measured velocity $v_x$ was smoothed using a 0.3-s moving-average window and denoted by the subscript $(.)_{\text{MA}}$.

trials show consistent qualitative and quantitative trends. The difference is particularly evident at 0.7 m/s: the proposed method achieves a 100% success rate, whereas the non-adaptive baseline succeeds in only one trial. The proposed method and vanilla RL both achieve a 100% success rate in simulation, while our framework achieves better performance in most reported metrics. Moreover, the proposed pipeline requires only approximately 12% of the total data-generation and training time of the RL baseline (Table I), providing a practical advantage when extending the framework to additional environmental or control parameters. Although the proposed adaptation can produce more conservative motion and slightly reduce traversal efficiency (Mean velocity in Table II), the evaluated results show that it reduces excessive slippage and falls by balancing velocity tracking with the contact-wrench-cone constraints.

A more detailed analysis of the online adaptation behavior is provided by the hardware results in Fig. 6. When a transition to the low-friction region is inferred, $\mu^{\text{wbc}}$ decreases from 1.1 to 0.4 in the 0.5 m/s trial, and from 0.6 to 0.4 in the 0.7 m/s trial as the robot moves from ① to ②. Over the same time interval, the locomotion velocity command is also adjusted, decreasing from 0.5 to 0.3 at 0.5 m/s and from 0.5 to 0.2 at 0.7 m/s, respectively. After the robot leaves the low-friction region, the control parameters increase above their minimum values, as shown at point ③. As shown in the second row of Fig. 6, the non-adaptive WBC exhibits substantial velocity-tracking degradation as the robot loses balance and falls. In contrast, under both velocity conditions, the measured locomotion velocity, shown by the blue solid line, closely follows the adaptively adjusted command generated by the proposed approach. These results indicate that the robot adaptively shifts toward more conservative control-parameter combinations when operating on slippery terrain.

This transition is further reflected in the evolution of the terrain-conditioned cost map shown in the last row of Fig. 6. Upon entering the slippery terrain, the framework continuously updates the WBC friction coefficient and commanded velocity based on the inferred friction, progressively shifting toward lower-cost parameter combinations associated with reduced slippage and successful traversal. By retaining the nominal WBC and adapting only physically meaningful parameters, the proposed method provides safer locomotion without replacing the underlying model-based controller.

## V. Conclusion

This paper presented LA-WBC, a learning-driven locomotion framework that integrates terrain-friction inference and online parameter adaptation with a nominal whole-body controller.

The framework improves sim-to-real friction inference through hardware-informed model matching and feature scaling, and adapts a low-dimensional set of physically meaningful parameters: the friction coefficient used by the WBC and the commanded locomotion velocity. Compared with the evaluated end-to-end RL baseline, the proposed training pipeline required only approximately $12\%$ of the total data-generation and training time. Complementary simulation and hardware experiments demonstrated improved locomotion performance over slippery terrain, including reduced slippage and body-motion variation and increased task-success rates relative to non-adaptive WBC.

Future work will extend the framework to non-rigid and time-varying terrain properties and to loco-manipulation tasks. We will also investigate adaptation across multiple conventional and non-conventional locomotion modes to support long-horizon operation in unstructured environments.

## REFERENCES


[1] K. Bouyarmane, K. Chappellet, J. Vaillant, and A. Kheddar, "Quadratic programming for multirobot and task-space force control," *IEEE Transactions on Robotics*, vol. 35, no. 1, pp. 64–77, 2019.

[2] C. D. Bellicoso, F. Jenelten, P. Fankhauser, C. Gehring, J. Hwangbo, and M. Hutter, "Dynamic locomotion and whole-body control for quadrupedal robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 3359–3365.

[3] D. Kim, S. J. Jorgensen, J. Lee, J. Ahn, J. Luo, and L. Sentis, "Dynamic locomotion for passive-ankle biped robots and humanoids using whole-body locomotion control," *The International Journal of Robotics Research*, vol. 39, no. 8, pp. 936–956, 2020.

[4] S. Caron, Q.-C. Pham, and Y. Nakamura, "Stability of surface contacts for humanoid robots: Closed-form formulae of the contact wrench cone for rectangular support areas," in *IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2015, pp. 5107–5112.

[5] C. Mastalli, I. Havoutis, M. Focchi, D. G. Caldwell, and C. Semini, "Motion planning for quadrupedal locomotion: Coupled planning, terrain mapping, and whole-body control," *IEEE Transactions on Robotics*, vol. 36, no. 6, pp. 1635–1648, 2020.

[6] M. Focchi, V. Barasuol, M. Frigerio, D. G. Caldwell, and C. Semini, *Slip Detection and Recovery for Quadruped Robots*. Cham: Springer International Publishing, 2018, pp. 185–199.

[7] M. Neunert, M. Stäuble, M. Giftthaler, C. D. Bellicoso, J. Carius, C. Gehring, M. Hutter, and J. Buchli, "Whole-body nonlinear model predictive control through contacts for quadrupeds," *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 1458–1465, 2018.

[8] G. Xin, W. Wolfslag, H.-C. Lin, C. Tiseo, and M. Mistry, "An optimization-based locomotion controller for quadruped robots leveraging Cartesian impedance control," *Frontiers in Robotics and AI*, vol. 7, p. 48, 2020.

[9] Y. Nisticò, S. Fahmi, L. Pallottino, C. Semini, and G. Fink, "On slip detection for quadruped robots," *Sensors*, vol. 22, no. 8, p. 2967, 2022.

[10] A. Agarwal, A. Kumar, J. Malik, and D. Pathak, "Legged locomotion in challenging terrains using egocentric vision," in *Proceedings of The 6th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, K. Liu, D. Kulic, and J. Ichnowski, Eds., vol. 205. PMLR, 2023, pp. 403–415.

[11] P. Ewen, A. Li, Y. Chen, S. Hong, and R. Vasudevan, "These maps are made for walking: Real-time terrain property estimation for mobile robots," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7083–7090, 2022.

[12] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.

[13] M. Brandao, K. Hashimoto, and A. Takanishi, "Friction from vision: A study of algorithmic and human performance with consequences for robot perception and teleoperation," in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2016, pp. 428–435.

[14] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.

[15] S. Choi, G. Ji, J. Park, H. Kim, J. Mun, J. H. Lee, and J. Hwangbo, "Learning quadrupedal locomotion on deformable terrain," *Science Robotics*, vol. 8, no. 74, p. eade2256, 2023.

[16] J. Tan, T. Zhang, E. Coumans, A. Iscen, Y. Bai, D. Hafner, S. Bohez, and V. Vanhoucke, "Sim-to-real: Learning agile locomotion for quadruped robots," in *Proceedings of Robotics: Science and Systems (RSS)*, 2018.

[17] G. B. Margolis, G. Yang, K. Paigwar, T. Chen, and P. Agrawal, "Rapid locomotion via reinforcement learning," *The International Journal of Robotics Research*, vol. 43, no. 4, pp. 572–587, 2024.

[18] V. Tsounis, M. Alge, J. Lee, F. Farshidian, and M. Hutter, "DeepGait: Planning and control of quadrupedal gaits using deep reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3699–3706, 2020.

[19] K.-C. Hsu, A. Z. Ren, D. P. Nguyen, A. Majumdar, and J. F. Fisac, "Sim-to-Lab-to-Real: Safe reinforcement learning with shielding and generalization guarantees," *Artificial Intelligence*, vol. 314, p. 103811, 2023.

[20] L. Smith, J. C. Kew, X. B. Peng, S. Ha, J. Tan, and S. Levine, "Legged robots that keep on learning: Fine-tuning locomotion policies in the real world," in *International Conference on Robotics and Automation (ICRA)*, 2022, pp. 1593–1599.

[21] P. Wu, W. Xie, J. Cao, H. Lai, and W. Zhang, "LoopSR: Looping sim-and-real for lifelong policy adaptation of legged robots," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 5649–5656.

[22] S. Gangapurwala, A. Mitchell, and I. Havoutis, "Guided constrained policy optimization for dynamic quadrupedal robot locomotion," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3642–3649, 2020.

[23] J. Choi, F. Castañeda, C. J. Tomlin, and K. Sreenath, "Reinforcement learning for safety-critical control under model uncertainty, using control Lyapunov functions and control barrier functions," in *Proceedings of Robotics: Science and Systems (RSS)*, 2020.

[24] D. Kang, J. Cheng, M. Zamora, F. Zargarbashi, and S. Coros, "RL + model-based control: Using on-demand optimal control to learn versatile legged locomotion," *IEEE Robotics and Automation Letters*, vol. 8, no. 10, pp. 6619–6626, 2023.

[25] P. Brakel, S. Bohez, L. Hasenclever, N. Heess, and K. Bousmalis, "Learning coordinated terrain-adaptive locomotion by imitating a centroidal dynamics planner," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 10 335–10 342.

[26] S. Gangapurwala, M. Geisert, R. Orsolino, M. Fallon, and I. Havoutis, "RLOC: Terrain-aware legged locomotion using reinforcement learning and optimal control," *IEEE Transactions on Robotics*, vol. 38, no. 5, pp. 2908–2927, 2022.

[27] ——, "Real-time trajectory adaptation for quadrupedal locomotion using deep reinforcement learning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 5973–5979.

[28] J. Ahn, J. Lee, and L. Sentis, "Data-efficient and safe learning for humanoid locomotion aided by a dynamic balancing model," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 4376–4383, 2020.

[29] W. Ubellacker and A. D. Ames, "Learned regions of attraction for safe motion primitive transitions," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 3643–3650.

[30] M. H. Raibert, *Legged Robots That Balance*. Cambridge, MA: The MIT Press, 1986.

[31] J. Lee, J. Ahn, D. Kim, S. H. Bang, and L. Sentis, "Online gain adaptation of whole-body control for legged robots with unknown disturbances," *Frontiers in Robotics and AI*, vol. 8, p. 788902, 2022.

[32] A. Albu-Schäffer, C. Ott, and G. Hirzinger, "A unified passivity-based control framework for position, torque and impedance control of flexible joint robots," *The International Journal of Robotics Research*, vol. 26, no. 1, pp. 23–39, 2007.

[33] S. Fahmi, C. Mastalli, M. Focchi, and C. Semini, "Passive whole-body control for quadruped robots: Experimental validation over challenging terrain," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2553–2560, 2019.

[34] DeepMind, "MuJoCo documentation — solimp," https://mujoco.readthedocs.io/en/stable/modeling.html#solimp, accessed: 2025-09-05.

[35] H. Lin, H. Li, and Y. Gao, "See-touch-predict: Active exploration and online perception of terrain physics with legged robots," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3470–3477, 2025.

[36] A. Jain, A. R. Zamir, S. Savarese, and A. Saxena, "Structural-RNN: Deep learning on spatio-temporal graphs," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, June 2016.

[37] B. Liang, L. Sun, X. Zhu, B. Zhang, Z. Xiong, Y. Wang, C. Li, K. Sreenath, and M. Tomizuka, "Adaptive energy regularization for autonomous gait transition and energy-efficient quadruped locomotion," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 5350–5356.

[38] Unitree Robotics. (2025) unitree_rl_lab. Accessed: 2026-07-17. [Online]. Available: https://github.com/unitreerobotics/unitree_rl_lab